

Vihaan Patel

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EDUCATION

Bachelors, Computer Systems Engineering (Honors)

May 2026

Arizona State University, Tempe, AZ

GPA 3.87

WORK EXPERIENCE

Research Intern (*Arizona State University*)

August 2025 – Present

- Built an agentic stimulus-generation workflow achieving 96-100% coverage across sub-system level RTL designs, reducing token consumption 4x over baseline by encoding verification methodology into tools, prompts, and context retrieval.
- Used trace-level analysis to diagnose uncovered verification paths and inconsistent stimulus generation, identifying design comprehension as the bottleneck and proposing a workflow with improved context management and retrieval.

Lead Tutor, Academic Support Network (*Arizona State University*)

January 2025 – May 2026

- Spearheaded the curriculum design and delivery of high-impact review sessions for CS, Math, and Physics, standardizing tutoring methodologies to ensure consistency across a network of 20,000+ students.
- Partnered with university faculty to execute campus-wide technical workshops, successfully scaling academic support operations and bridging the communication gap between course requirements and student success.

RELEVANT PROJECTS

RTL PPA Optimization Agent and Evaluation (PPAgent)

Spring 2026

- Built an RTL optimization benchmark on picorv32 RISC-V core modules with a two-stage evaluation harness (Icarus Verilog correctness simulation, Yosys PPA synthesis) that scores on cell/flip-flop/wire count deltas against baseline.
- Extended mini-swe-agent with simulate/synthesize tools and evaluated it against manual optimizations.

Asynchronous FIFO: CDC Design + DPI Golden-Model Verification

Summer 2025

- Designed a parameterized asynchronous FIFO in SystemVerilog with Gray-code pointers, dual-flop CDC synchronizers, and almost-full/empty flags, operating across independent read and write clock domains.
- Verified with a C/C++ golden reference model bound via SystemVerilog DPI and a class-based scoreboard (monitor, mailbox, queue model); plusarg bug injection and SVA confirmed checkers catch data and flag mismatches.

Pipelined MIPS Processor: Constrained-Random Stimulus + UVM Functional Coverage

Summer 2025

- Built a constrained-random instruction generator emitting legal ADD/AND/LW/SW/BEQ streams with RAW register hazards, store-load dependencies, branch taken/not-taken, and randomized gaps.
- Modeled UVM functional coverage: a uvm_monitor publishes fetched instructions to a uvm_subscriber whose covergroups bin fields, dependencies, and gaps; merged multi-seed runs with URG to close coverage near 100%.

Systolic Matrix-Multiply Accelerator: AMBA APB Design + UVM Verification

Summer 2025

- Designed a 4x4 systolic matrix-multiply accelerator in SystemVerilog with FP8 MAC units and an AMBA APB slave exposing a CSR register map (operand addresses, strides, start, done); synthesized in Synopsys Design Compiler.
- Verified in UVM: APB master VIP agent programs CSRs and polls done; custom memory-slave agents (uvm_config_db) model A/B/C SRAMs; virtual sequence drives a self-checking scoreboard with SVA and force/release fault injection.

TECHNICAL SKILLS

- **Programming:** Python, C++, Java, Bash/Shell, SQL
- **DV Methodology:** UVM, SystemVerilog, SVA, functional coverage, constrained-random verification, DPI golden models
- **EDA Tools:** Synopsys VCS, Verdi, Design Compiler, URG, QuestaSim, Vivado
- **Machine Learning:** PyTorch, Scikit-learn, NumPy, TensorFlow/Keras, Reinforcement Learning
- **GenAI:** LangChain, LangGraph, HuggingFace, OpenAI API, RAG, MCP, Context Engineering
- **Infrastructure:** vLLM, CUDA, RAPIDS, TensorRT, HPC/SLURM, Linux, Docker, AWS EC2, Git, CI/CD